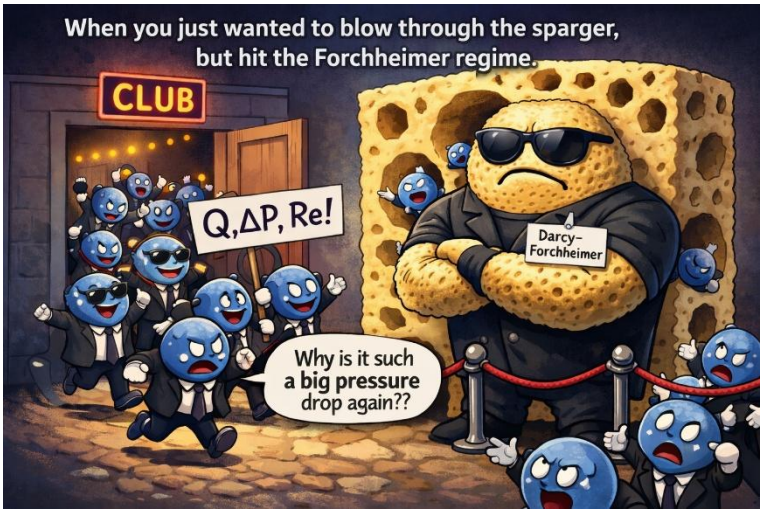
Gas Flow Through a Porous Medium

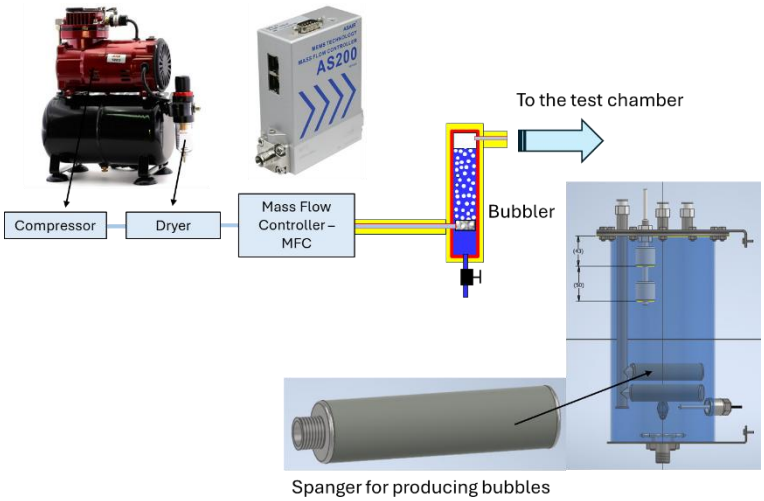


Contents

Introduction 2
Porous Materials 3
Manufacturing Technologies for Porous Materials and Their Key Parameters 3
Gas Flow Through a Porous Material 4
Accounting for Gas Compressibility 13
Numerical Simulation of Gas Pressure Drop Across a Porous Material 16

Introduction

Our interest in this topic arose in connection with the development of a bubbler for air humidification in a test heating chamber used for measuring the specific electrical resistivity of dust layers under high voltage. The main components of a bubbler are as follows: a flask containing distilled water; a porous sparger element immersed in the water and producing fine bubbles; and a compressor supplying air to the porous element. As the bubbles pass through the water layer heated to a specified temperature, they become humidified to nearly 100% RH and carry moisture into the outlet pipe, which delivers the humidified air to the test chamber. To prevent condensation, both the pipe itself and the test chamber must be maintained at a temperature higher than the water temperature in the sparger. The target parameters of the system are the volumetric humidity in the test chamber, which unlike relative humidity does not change with increasing temperature, and the gas exchange rate within the chamber.



The design of such a system requires the ability to estimate the pressure drop across the sparger, which is usually cylindrical in shape, as a function of gas flow rate, either at the surface of the sparger element itself or inside the test chamber. In our case, the test chamber connected to the sparger outlet operates at a pressure close to atmospheric, possibly slightly above it. If the hydrostatic pressure acting on the porous element and the surface-tension effects associated with bubble formation at its surface are neglected, the gas discharge from the porous element may be treated approximately as a free outflow into a cavity at atmospheric pressure.

The theoretical framework developed in this report is well presented in the published literature. We have selected only a few references that illustrate these approaches:

- Materials 2021, 14(12), 3153; https://doi.org/10.3390/ma14123153
- Metals 2015, 5(1), 336-349; https://doi.org/10.3390/met5010336
- Energies 2022, 15(3), 1197; https://doi.org/10.3390/en15031197
- Appl. Sci. 2018, 8(9), 1670; https://doi.org/10.3390/app8091670

To calculate the pressure drop across porous samples, we have developed our own Python programs, which are freely available for download from GitHub and may be used for any purpose without restriction.

Porous Materials

Manufacturing Technologies for Porous Materials and Their Key Parameters

The Powder Metallurgy Federation report Porous Metal Design Guidebook describes the following manufacturing technologies for porous media:

1) **Axial Compaction and Sintering.** In this approach, metal powder is pressed in a rigid die at a relatively low pressure sufficient to produce a "green" part, after which the compact is sintered at a temperature below the melting point of the metal. For porous parts, compared with conventional powder-metallurgy structural components, lower compaction pressures and an appropriate powder selection are used in order to preserve the required pore openness. The report identifies this technology as typical for stainless steel, titanium, nickel, nickel alloys, and part of bronze products; its advantages include high productivity, good permeability control, and good dimensional repeatability.
2) **Gravity Sintering, sometimes described as loose powder sintering.** Here the powder is poured into the mould cavity without external pressing and then sintered so that metallurgical necks form between particles. This method is particularly typical for bronze and is well suited for relatively inexpensive porous parts. Draft angles, tolerances, and shape limitations are discussed for this process; for example, a dimensional tolerance of approximately ±2% is cited as a guideline, together with a looser tolerance in length.
3) **Powder Rolling and Sintering.** This is the case where the report provides the most useful numerical guidance on thickness. It states that porous P/M sheet can be made from stainless steel, copper, bronze, nickel alloys, and titanium either by direct rolling of powder or by powder deposition followed by calendaring prior to sintering. For this material, the report explicitly indicates that, depending on density and material, sheet thickness ranges from 0.25 mm to 3 mm. This is the clearest range of "typical thicknesses" of a porous layer that can be extracted from the document. Moreover, such sheet can be cut, bent, rolled, and welded into other configurations.
4) **Isostatic Compaction and Sintering.** Here pressure is applied not uniaxially but more uniformly through a deformable shell filled with powder. This method is particularly useful for parts with a large length-to-diameter ratio, including tubes and special shapes, and can be applied to virtually all common P/M materials. The section on dimensional limitations also states that isostatic compaction allows parts starting from 1/8 inch in diameter, with a typical diameter of about 3 inches, while lengths of up to 72 inches can be achieved by welding sections together.
5) **Metal Spraying.** Molten metal is sprayed onto a substrate, while porosity is controlled either by the spraying conditions or by co-spraying a second material that is later removed. This is no longer a classical bulk powder compact, but rather the formation of a porous structure on a substrate.
6) **Metal Coating and Sintering.** In this case, metal powder is mixed with a binder to obtain a slurry, which is then applied to a porous substrate or used to form a near-net-shape preform. The document specifically emphasizes the need for careful binder removal and uniform porosity control.

7) **Metal Injection Molding and Sintering (MIM).** This route is described as mixing powder with a substantial amount of special binders to obtain a viscous feedstock suitable for injection moulding. After binder removal and sintering, parts with a controlled density are obtained. This approach requires specialized equipment because of the large shrinkage during debinding.

Porosity, pore size, and permeability are related but distinct characteristics of a porous medium. **Porosity** ε shows what fraction of the material volume is occupied by voids: $\varepsilon = \frac{V_{\text{pore}}}{V_{\text{total}}}$, where V_{pore} is the pore volume and V_{total} is the total material volume. In other words, porosity answers the question of how much empty space exists in the material as a whole. **Pore size** characterizes the scale of individual pore channels or throats. It is usually denoted by d_p and is often given in micrometers. Thus, pore size answers the question of how large or small the passages in the porous structure are. It is important to remember, however, that in a real sintered material the pores are not identical; therefore, "pore size" usually means not the literal size of every pore, but rather a nominal or effective value defined according to the manufacturer's methodology.

Gas Flow Through a Porous Material

Let V_c be the internal volume of the test chamber (m^3), n — the atmosphere renewal rate (s^{-1}), and Q_c — the volumetric gas flow rate in the test chamber (m^3/s). Then:

$$Q_c = nV_c \tag{1}$$

This is the gas flow rate under the thermal chamber conditions, i.e. at its operating temperature and nearly atmospheric pressure.

Let T_c be the absolute gas temperature in the thermal chamber (K), P_c — the absolute gas pressure in the thermal chamber (Pa), Q_s — the volumetric gas flow rate at the porous sparger surface, i.e. in the pore-exit zone (m^3/s), T_s — the gas temperature at the sparger surface (K), and P_s — the absolute gas pressure at the pore outlet (Pa). Since the gas subsequently enters the thermal chamber after the bubbler and is heated there, the flow rate in the chamber and the flow rate at the sparger surface are not the same. They are related by the ideal-gas equation for the same mass flow rate:

$$\frac{P_s Q_s}{T_s} = \frac{P_c Q_c}{T_c} \tag{2}$$

Hence:

$$\boxed{Q_s = Q_c \frac{P_c}{P_s} \frac{T_s}{T_c}} \tag{3}$$

Equation (3) is the first key relation: the flow rate at the sparger surface is determined from the flow rate in the thermal chamber. In our case, P_c and P_s are close to atmospheric pressure.

Let P_{head} be the absolute gas pressure above the liquid in the bubbler (Pa), ρ_ℓ — the liquid density (kg/m^3), g — the gravitational acceleration (m/s^2), and h — the immersion depth of the sparger element (m). Then the liquid pressure at the outer sparger surface is:

$$P_b = P_{\text{head}} + \rho_\ell g h \tag{4}$$

where P_{head} is close to atmospheric pressure in our setup. For a shallow immersion depth, P_b exceeds atmospheric pressure only slightly.

Let σ be the liquid surface tension (N/m), θ — the wetting contact angle, d_p — the characteristic pore size (m), and ΔP_{cap} — the capillary pressure drop (Pa). Then:

$$\Delta P_{\text{cap}} \approx \frac{4\sigma\cos\,\theta}{d_p} \tag{5}$$

This is the additional pressure required for the gas to displace the meniscus from the pore and begin forming bubbles.

The gas at the sparger surface must overcome the liquid pressure and the capillary threshold, therefore:

$$P_s \approx P_b + \Delta P_{\text{cap}} \tag{6}$$

Thus, P_s is no longer just atmospheric pressure, but atmospheric pressure plus the hydrostatic and capillary contributions. This value is substituted into (3).

Let A_p be the active porous surface area of the sparger (m^2), and u_s — the superficial gas velocity through the sparger (m/s). Then:

$$A_p = \frac{Q_s}{u_s} \tag{7}$$

By active porous surface area we mean the part of the sparger surface through which gas actually exits into the liquid. It is at this stage that the sparger geometry is selected. If the allowable superficial velocity u_s is specified, the required area A_p can be found immediately.

Let us develop a phenomenological theory of gas flow through flat and cylindrical porous layers and derive the pressure-drop equations in the layer. In a porous medium, the local one-dimensional form of the Darcy–Forchheimer equation is written as:

$$-\frac{dP}{dx} = \frac{\mu_g}{k}u + \beta\rho_g u^2 \tag{8}$$

where

P is pressure (Pa),

x is the coordinate along the flow direction (m),

μ_g is the gas dynamic viscosity (Pa · s),

k is the permeability of the porous medium (m^2),

β is the inertial coefficient (1/m),

ρ_g is the gas density (kg/m^3),

u is the filtration velocity (m/s).

The first term $\frac{\mu_g}{k}$ corresponds to Darcy viscous losses, while the second term $u\beta\rho_g u^2$ represents Forchheimer inertial losses. In some papers, the second term in (8) is written with the coefficient $\frac{\beta\rho_g}{\sqrt{k}}$, in which case β becomes a dimensionless parameter. The nonlinear term should use the local gas density ρ_g in the porous medium. Because the pressure decreases across the thickness of the porous material, the density also changes. In a rigorous formulation, the quadratic term should account for the dependence of density on the local pressure. However, in engineering practice it is acceptable to use a representative average density corresponding to the mean absolute pressure in the porous wall. If a conservative estimate is required, the inlet gas density may be used, which leads to some overestimation of the inertial contribution. A more accurate treatment will be developed later.

The inertial coefficient in the Darcy–Forchheimer equation is needed to describe the portion of the pressure loss that is no longer associated solely with viscous friction against the pore walls, but rather arises from accelerations, flow rearrangements, and local vortical effects in a complex porous structure.

Permeability is a property of a porous medium that indicates how easily a gas or liquid passes through it under a pressure gradient. **Dynamic viscosity** is a property of the gas or liquid itself that indicates how strongly the medium resists shear and flow. Physically, this can be interpreted as follows: higher viscosity means the medium flows “more heavily”, while lower viscosity means it flows “more easily”. For example, air has a low viscosity, water has a higher viscosity, and oil has an even higher viscosity. In the context of a bubbler, the gas dynamic viscosity describes how strongly the gas itself resists flow through narrow pore channels. Therefore, all else being equal, the larger μ, the greater the pressure drop; the larger k, the smaller the pressure drop.

Two media may have the same porosity ε but different permeability k. In that case, the void fraction is the same, but gas will flow through them differently. Likewise, two media may have the same nominal pore size but different permeability if one has a more connected and less tortuous structure. The relation between ε and k may be roughly expressed by a Kozeny–Carman-type equation:

$$k \sim \frac{\varepsilon^3}{(1-\varepsilon)^2} d_p^2 \tag{9}$$

where d_p is the characteristic pore size. This equation is not exact for any arbitrary sintered material, but it clearly shows the following trends:

- with increasing porosity, permeability usually increases
- with increasing pore size, permeability also increases
- for fine pores and low porosity, permeability decreases sharply

For a bubbler, this means the following:

- fine bubbles require finer pores
- finer pores usually reduce permeability
- reduced permeability increases the pressure drop

Thus, in a bubbler there is always a compromise between bubble dispersion and pressure resistance.

For a flat wall of thickness L and constant filtration area A, with volumetric flow rate Q, the filtration velocity is constant and equal to:

$$u = \frac{Q}{A} \tag{10}$$

In some studies, the filtration velocity is defined as $u = \frac{Q}{A\varepsilon}$, where ε is the porosity. Division by porosity is associated with the fact that the flow velocity is determined not from the total geometric cross-sectional area of the specimen, but from the effective void cross-sectional area estimated as $A\varepsilon$. Therefore, the velocity used is the interstitial velocity rather than the superficial velocity. In works where the velocity is defined with respect to the full sample area, the coefficient ε is absent from the velocity formula and its effect is effectively absorbed into the fitted coefficients of the model. Substituting $u = \frac{Q}{A}$ into equation (8):

$$-\frac{dP}{dx} = \frac{\mu_g}{k}\frac{Q}{A} + \beta\rho_g\left(\frac{Q}{A}\right)^2 \tag{11}$$

Hence:

$$dP = -\left[\frac{\mu_g}{k}\frac{Q}{A} + \beta\rho_g\left(\frac{Q}{A}\right)^2\right]dx \tag{12}$$

Integrating across the thickness from $x = 0$ to $x = L$, we obtain the pressure drop across the flat wall:

$$\Delta P_{\text{flat}} = \frac{\mu_g}{k}qL + \beta\rho_g L q^2 \tag{13}$$

where $q = \frac{Q}{A}$ is the specific flow rate.

Now consider a cylindrical bubbler of length L_c with inner radius R_{in} and outer radius R_{out}. The gas flows radially, so the filtration area depends on radius:

$$A(r) = 2\pi r L_c \tag{14}$$

For the total flow rate Q, the local filtration velocity at radius r is:

$$u(r) = \frac{Q}{2\pi r L_c} \tag{15}$$

The local radial form of equation (8) becomes:

$$-\frac{dP}{dr} = \frac{\mu_g}{k}u(r) + \beta\rho_g u(r)^2 \tag{16}$$

Substituting $u(r)$:

$$-\frac{dP}{dr} = \frac{\mu_g}{k}\frac{Q}{2\pi r L_c} + \beta\rho_g\left(\frac{Q}{2\pi r L_c}\right)^2 \tag{17}$$

Hence:

$$dP = -\left[\frac{\mu_g Q}{2\pi k L_c}\frac{1}{r} + \beta\rho_g\frac{Q^2}{4\pi^2 L_c^2}\frac{1}{r^2}\right]dr \tag{18}$$

Integrating with respect to radius from R_{in} to R_{out}, we obtain the pressure drop across the cylindrical wall:

$$\boxed{\Delta P_{\text{cyl}} = \frac{\mu_g Q}{2\pi k L_c}\ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \beta\rho_g\frac{Q^2}{4\pi^2 L_c^2}\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right)} \tag{19}$$

For a bubbler, it is convenient to introduce the outer surface area $A_{\text{out}} = 2\pi R_{\text{out}} L_c$ and the specific gas flow rate referred to the outer surface $q_{\text{out}} = \frac{Q_s}{A_{\text{out}}}$. Then:

$$Q_s = q_{\text{out}}\, 2\pi R_{\text{out}} L_c \tag{20}$$

Substituting this into the cylindrical formula:

$$\boxed{\Delta P_{\text{cyl}} = \frac{\mu_g}{k} q_{\text{out}} R_{\text{out}} \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \beta\rho_g q_{\text{out}}^2 R_{\text{out}}^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right)} \tag{21}$$

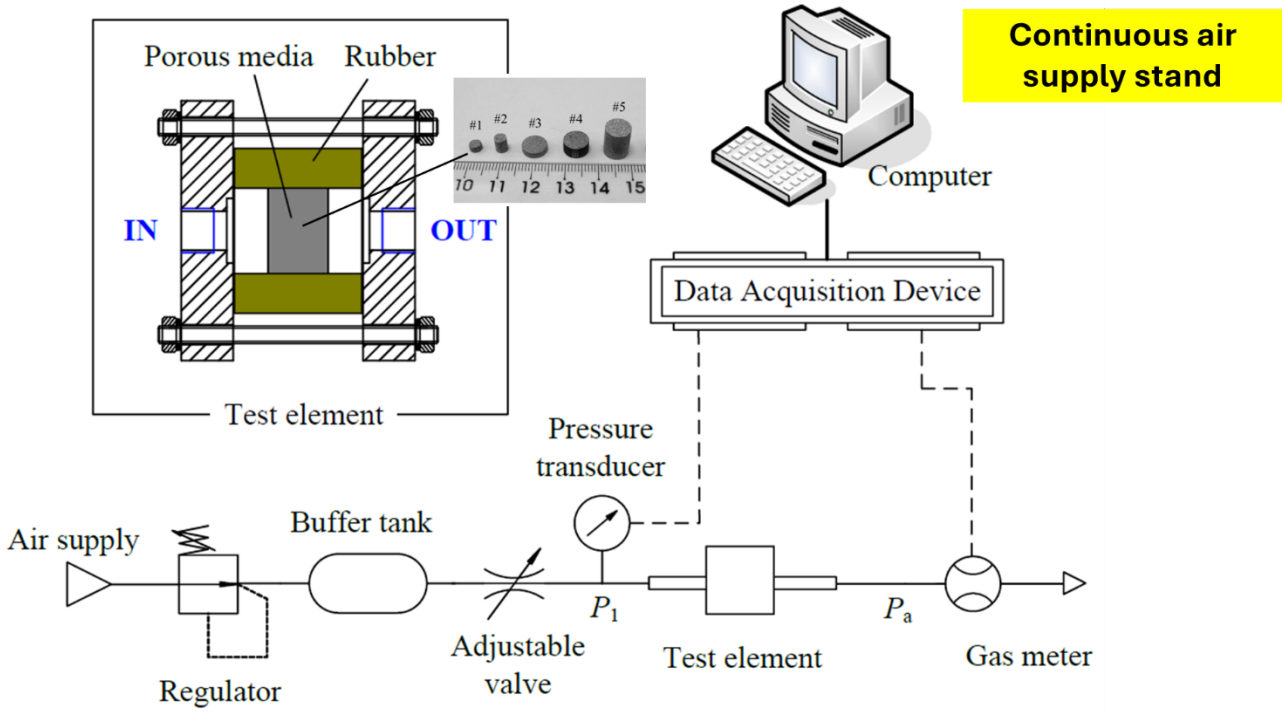
At low filtration velocities, only the linear terms may be retained:

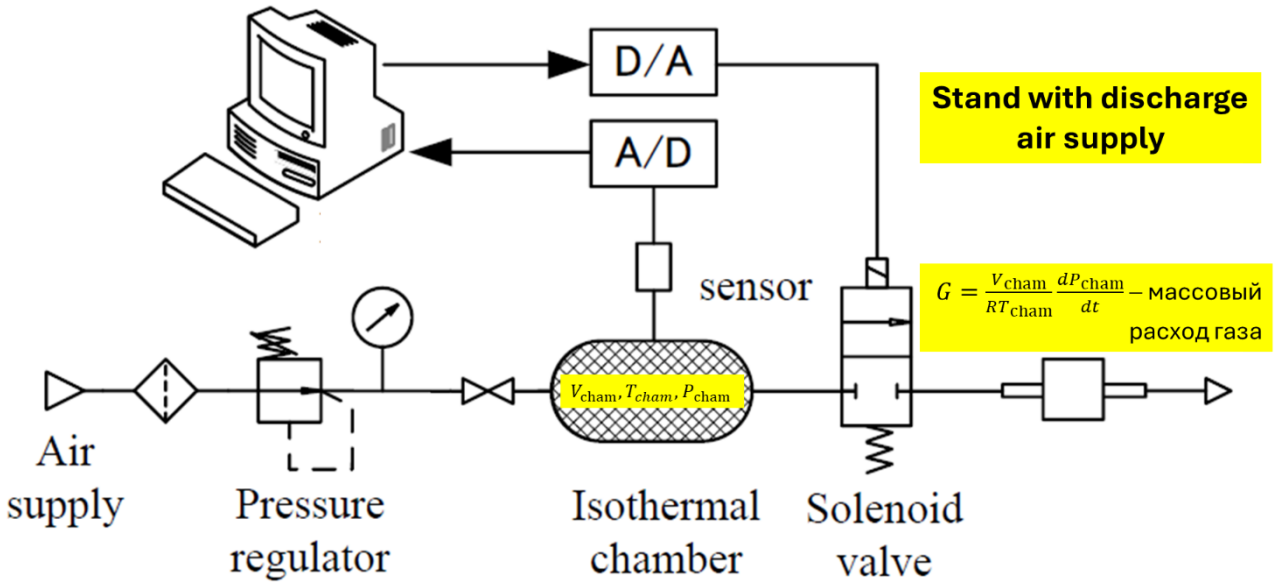
$$\Delta P_{\text{flat}} \approx \frac{\mu_g}{k} qL \tag{22}$$

$$\Delta P_{\text{cyl}} \approx \frac{\mu_g}{k} q_{\text{out}} R_{\text{out}} \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) \tag{23}$$

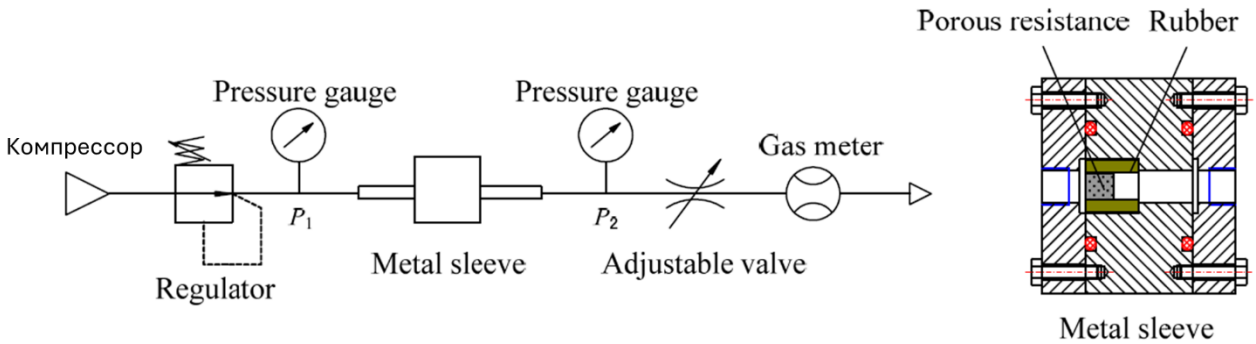
If μ_g is essentially a material parameter of the flowing medium itself (gas or liquid), then k strongly depends on the sintering technology of the porous material. Both parameters must therefore be determined experimentally.

Experimental determination of the constants in the Darcy–Forchheimer equation for a specific liquid/gas and porous material can be performed on flat specimens ("pellets") using the test rigs shown in the figures below, where the outlet pressure is close to atmospheric (free discharge).

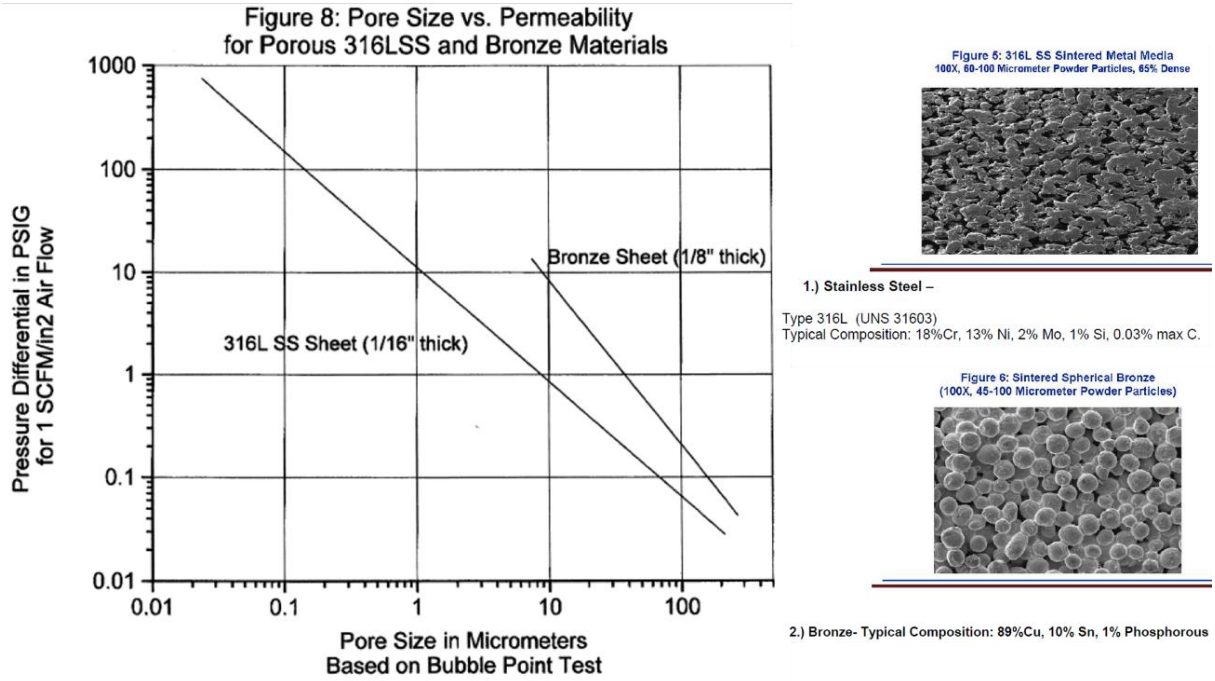




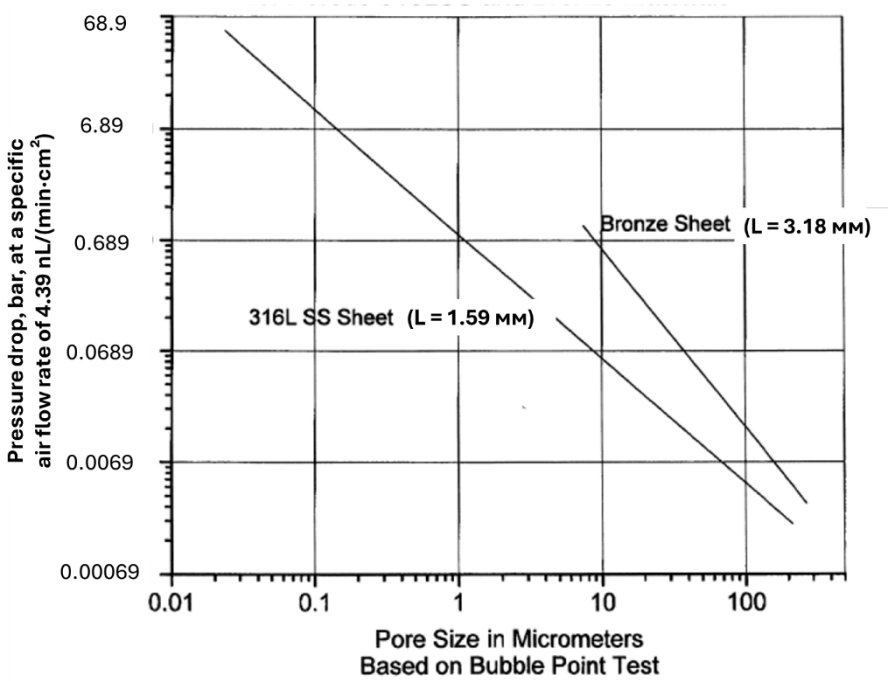
In another type of test rig, shown in the figure below, the pressure P_1 upstream of the specimen is kept constant, while the pressure difference is controlled by the outlet pressure P_2.



In the Porous Metal Design Guidebook, pressure-drop curves are presented, see the figure below, for flat porous specimens ("pellets") as a function of pore size d_p at a fixed specific gas flow rate q for two sintered materials: stainless steel and bronze.



An interpretation of this plot in units convenient for our purposes is shown below.



For fixed d_p, the ratio $\frac{\mu_g}{k(d_p)}$ can be calculated from the experimental plot values:

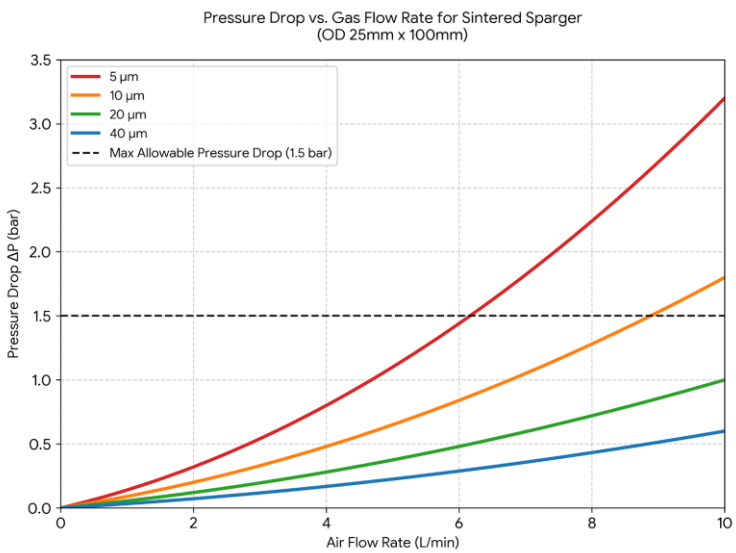
$$\frac{\mu_g}{k(d_p)} = \frac{\Delta P_{\text{flat}}(d_p)}{qL} \tag{24}$$

Substituting (24) into (23), we obtain a conversion formula from the flat geometry to the cylindrical geometry most commonly used in practice:

$$\Delta P_{\text{cyl}} \approx \Delta P_{\text{flat}}(d_p) \frac{R_{\text{out}}}{L} \frac{q_{\text{out}}}{q} \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) \tag{25}$$

In the plot above, the layer thickness is $L = 1.59$ mm for stainless steel and $L = 3.18$ mm for bronze. The specific gas flow rate $q = 4.39$ nL/(min·cm^2) at atmospheric discharge from the specimen refers to normal litres (nL). We use these values to extrapolate to a cylindrical sparger element with the following geometric parameters: $d_p = 10$ μm; $R_{\text{out}} = 1.25$ cm; $R_{\text{in}} = R_{\text{out}} - \Delta R = 1$ cm, where the cylindrical layer thickness is $\Delta R = 0.25$ cm and the cylinder length is $L_c = 10$ cm. At a gas flow rate of $Q_s = 10$ nL/min, the corresponding specific value is much lower than q: $q_{\text{out}} = \frac{Q_s}{2\pi R_{\text{out}} L_c} \approx 0.127$ nL/(min·cm^2). The final result is: $\Delta P_{\text{cyl}} \approx \Delta P_{\text{flat}}(10) \frac{R_{\text{out}}}{L} \frac{q_{\text{out}}}{q} \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) = 0.0689 \times \frac{1.25}{0.159} \times \frac{0.127}{4.39} \times \ln\left(\frac{1.25}{1}\right) \approx 0.0035$ bar.

For our bubbler, we plan to use cylindrical porous elements made of sintered 316L stainless steel and supplied by the company Saifilter. The geometric dimensions of the elements will be the same as in the example just considered. The company provided calculated dependences $\Delta P_{\text{cyl}}(Q_s)$ for various values of d_p, shown in the figure below. According to these data, at $Q_s = 10$ nL/min and $d_p = 10$ μm, the pressure drop is about 1.75 bar, which is approximately 500 times larger than the previously obtained estimate of 0.0035 bar.



The geometry of the sparger element selected for our bubbler is as follows: length 100 mm, diameter 25 mm, pore size 10 μm, thread 1/4 inch. We limit the maximum allowable pressure drop to 1.5 bar.

Such a sharp discrepancy (by a factor of 500) indicates that the simplified linear Darcy model used above—without the second term in (13) and (21) — is insufficient to describe the actual product. However, it should be noted that those equations were derived without taking gas compressibility into account. The flow–pressure curves provided by the manufacturer may have been constructed not as a direct solution of the compressible-flow model through a porous medium (see the next section), but rather as an engineering approximation of experimental data. In that case, coefficients obtained from testing may have been substituted into a simpler relation of the form $\Delta P = aQ + bQ^2$ without explicit compressibility. Such curves may satisfactorily describe the operating range of a particular product while differing in shape from the dependences obtained in the literature on the basis of a modified Forchheimer equation for gases. In particular, the work of Zhong et al. achieved good agreement with experiment precisely within a model that accounts for air compressibility. In the next section, we develop the theory with compressibility included.

The question of what low velocity may serve as the boundary between the viscous regime and the inertial regime remains open. In practice, it is very difficult to define a universal regime-transition boundary applicable to all types of porous media. Although specific threshold Reynolds numbers have been proposed by a number of researchers, these values depend on the properties of the porous materials and working fluids used. The nominal pore size d_p alone does not uniquely determine the hydraulic resistance, because the actual permeability is also governed by porosity, channel tortuosity, connectivity, and local constrictions. Consequently, even with the same nominal d_p, the actual permeability of a sintered material may be substantially lower than assumed in a preliminary estimate.

If a characteristic $\Delta P_{\text{por}}(Q_s)$ is available, calculated with gas compressibility included for a specific sparger geometry using experimentally determined hydrodynamic parameters, then the pressure drop across the porous medium may be written as:

$$P_1 - P_{\text{head}} = \rho_\ell g h + \frac{4\sigma\cos\theta}{d_p} + \Delta P_{\text{por}}(Q_s) + \Delta P_{\text{line}} \tag{26}$$

$$Q_s = Q_c \frac{P_c}{P_{\text{head}} + \rho_\ell g h + \frac{4\sigma\cos\theta}{d_p}} \frac{T_s}{T_c} \tag{27}$$

where ΔP_{line} is the pressure drop in the supply pipes and Q_c is the specified gas flow rate in the chamber. In most practical situations, when the sparger is immersed only shallowly in water and the pressure in the heating chamber is close to atmospheric, $P_1 - P_{\text{atm}} \approx \Delta P_{\text{por}}(Q_s)$ and $Q_s \approx Q_c \frac{T_s}{T_c}$, where ΔP_{line} has also been neglected.

Accounting for Gas Compressibility

In this section we develop a more general theory including gas compressibility. When the pressure changes, the local volumetric flow rate changes, but the mass is conserved. If P_1, Q_1 are specified at the porous-medium inlet, then at any point inside the medium:

$$\rho_g Q = \rho_1 Q_1 \tag{28}$$

where Q is the local volumetric flow rate under the current conditions (m^3/s) and ρ_1 is the gas density at the inlet. For a metallic porous sparger element immersed in water, the gas temperature inside the porous wall may, in a first approximation, be taken equal to the temperature of the sparger itself and of the surrounding water. This assumption is based on the high thermal conductivity of sintered metal, the large specific contact area between the gas and the pore walls, and the small thickness of the porous wall. Deviations from this approximation are possible at very high gas flow rates or when the temperature difference between the inlet gas and the water is large. Since for an ideal gas $\rho = \frac{P}{RT}$, at constant sparger temperature T_s and inlet-gas temperature T_1 we obtain $\frac{P}{RT_s} Q = \frac{P_1}{RT_1} Q_1$, and after cancellation:

$$Q = Q_1 \frac{T_s P_1}{T_1 P} \tag{29}$$

Let L be the thickness of the flat wall (m) and A the constant filtration area (m^2), then the local filtration velocity is:

$$u = \frac{Q}{A} = Q_1 \frac{T_s P_1}{T_1 P A} \tag{30}$$

Substituting (30) and $\rho_g = \frac{P}{RT_s}$ into the Darcy–Forchheimer equation and performing straightforward algebraic transformations, we obtain:

$$-\frac{1}{2}\frac{d(P^2)}{dx} = \frac{\mu_g T_s P_1}{k T_1 A} Q_1 + \frac{\beta T_s P_1^2}{R T_1^2 A^2} Q_1^2 \tag{31}$$

Integrating from the inlet $x = 0$, where the pressure is P_1, to the outlet $x = L$, where the pressure is P_2, we obtain:

$$\boxed{P_1^2 - P_2^2 = 2L\left(\frac{\mu_g T_s P_1}{k T_1 A} Q_1 + \frac{\beta T_s P_1^2}{R T_1^2 A^2} Q_1^2\right)} \tag{32}$$

This is a quadratic equation with respect to P_1.

Now consider a cylindrical sparger element. Let R_{in} be the inner radius (m), R_{out} — the outer radius (m), and L_c — the length of the cylindrical porous section (m). At radius r, the flow area is $A(r) = 2\pi r L_c$. Then the local velocity is $u(r) = \frac{Q}{2\pi r L_c}$. In accordance with (29), the local volumetric flow rate is again related to the inlet value through compressibility: $Q(r) = Q_1 \frac{T_s}{T_1}\frac{P_1}{P}$. Consequently, $u(r) = \frac{Q_1 T_s P_1}{2\pi r L_c T_1 P}$. Substituting into the local radial form of the Darcy–Forchheimer equation and performing straightforward algebra, we obtain:

$$-P\frac{dP}{dr} = \frac{\mu_g Q_1 T_s P_1}{2\pi k L_c T_1}\frac{1}{r} + \frac{\beta T_s Q_1^2 P_1^2}{2\pi^2 R L_c^2 T_1^2}\frac{1}{r^2} \tag{33}$$

Integrating both sides of (33) from $r = R_{\text{in}}$, where the pressure is P_1, to $r = R_{\text{out}}$, where the pressure is P_2:

$$P_1^2 - P_2^2 = \frac{\mu_g Q_1 T_s P_1}{2\pi k L_c T_1}\ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta T_s Q_1^2 P_1^2}{2\pi^2 R L_c^2 T_1^2}\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right) \tag{34}$$

This is a quadratic equation with respect to P_1.

The free flow rate Q_s leaving the bubbler enters the heating chamber with temperature T_c, pressure $P_2 \approx P_{atm}$, and flow rate Q_2. Let us express Q_1 in terms of Q_s, neglecting the hydrostatic pressure above the sparger element:

$$Q_1 = Q_s \frac{T_1 P_{atm}}{T_s P_1} \tag{35}$$

Here Q_s denotes the free volumetric gas flow rate immediately after it exits the sparger, while Q_c is the volumetric flow rate of the same gas in the heating chamber after heating to temperature T_c. Substituting (35) into (32) and (34), we obtain:

$$P_1^2 - P_{atm}^2 = 2L\left(\frac{\mu_g P_{atm}}{kA}Q_s + \frac{\beta P_{atm}^2}{R T_s A^2}Q_s^2\right) \tag{36}$$

$$P_1^2 - P_{atm}^2 = \frac{\mu_g P_{atm}}{2\pi k L_c}Q_s \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta P_{atm}^2}{2\pi^2 R T_s L_c^2}Q_s^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right) \tag{37}$$

The pressure P_1 can be expressed explicitly in both equations (36) and (37). Then, for the differences , we obtain:$\Delta P = P_1 - P_{atm}$

$$\Delta P_{\text{flat}}(Q_s) = \sqrt{P_{atm}^2 + 2L\left(\frac{\mu_g P_{atm}}{kA}Q_s + \frac{\beta P_{atm}^2}{R T_s A^2}Q_s^2\right)} - P_{atm} \tag{38}$$

$$\Delta P_{\text{cyl}}(Q_s) = \sqrt{P_{atm}^2 + \frac{\mu_g P_{atm}}{\pi k L_c}Q_s \ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta P_{atm}^2}{2\pi^2 L_c^2 R T_s}Q_s^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right)} - P_{atm} \tag{39}$$

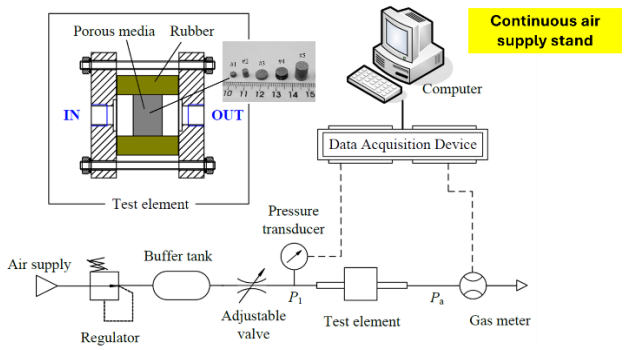
The pressure difference may also be expressed through the flow rate in the heating chamber Q_c, used as the target parameter, with $Q_s = \frac{T_s}{T_c}Q_c$ (the pressure in that part is approximately atmospheric).

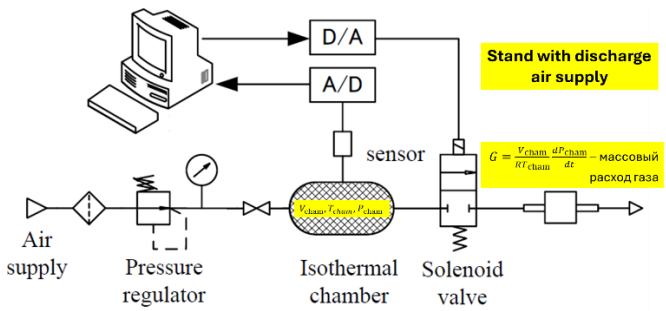
In many papers on gas flow through porous materials, the pressure difference is represented as a function of the mass flow rate $G = \frac{P}{RT}Q$. Therefore, the corresponding $Q = \frac{RT}{P}G$ must be expressed through the mass gas flow rate. Equations (38) and (39) then take the following form for $Q_s = \frac{RT_s}{P_{atm}}G$:

$$\Delta P_{\text{flat}}(G) = \sqrt{P_{atm}^2 + 2L\left(\frac{\mu_g RT_s}{kA}G + \frac{\beta RT_s}{A^2}G^2\right)} - P_{atm} \tag{40}$$

$$\Delta P_{\text{cyl}}(G) = \sqrt{P_{atm}^2 + \frac{\mu_g RT_s}{\pi k L_c}G\ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta RT_s}{2\pi^2 L_c^2}G^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right)} - P_{atm} \tag{41}$$

Equations (38)–(41) correspond to the measurement schemes shown in the figures below, where the outlet pressure is approximately atmospheric.

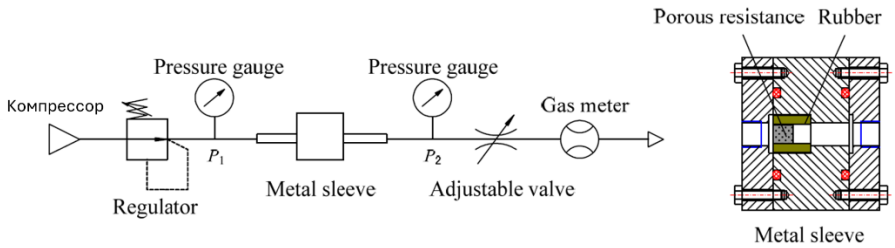




However, in the measurement scheme where the inlet pressure P_1 is maintained constant while the outlet pressure P_2 is varied, one has to return to the original equations:

$$P_1^2 - P_2^2 = 2L\left(\frac{\mu_g P_2}{kA}Q_s + \frac{\beta P_2^2}{RT_s A^2}Q_s^2\right) \tag{42}$$

$$P_1^2 - P_2^2 = \frac{\mu_g P_2}{2\pi k L_c}Q_s\ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta P_2^2}{2\pi^2 RT_s L_c^2}Q_s^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right) \tag{43}$$



Or, replacing $Q_s = \frac{RT_s}{P_2}G$:

$$P_1^2 - P_2^2 = 2L\left(\frac{\mu_g RT_s}{kA}G + \frac{\beta RT_s}{A^2}G^2\right) \tag{44}$$

$$P_1^2 - P_2^2 = \frac{\mu_g RT_s}{2\pi k L_c}G\ln\left(\frac{R_{\text{out}}}{R_{\text{in}}}\right) + \frac{\beta RT_s}{2\pi^2 L_c^2}G^2\left(\frac{1}{R_{\text{in}}} - \frac{1}{R_{\text{out}}}\right) \tag{45}$$

For this measurement scheme, one should first express P_2 through $P_1 = const$, and then calculate the difference $\Delta P = P_2 - P_1$.

Numerical Simulation of Gas Pressure Drop Across a Porous Material

Python programs for calculating the pressure drop in flat and cylindrical specimens for different measurement schemes are collected in the following folders:

- Python_Cylindrical_ sponge_dP(Qs)
- Python_Cylindrical_sponge_dP(G)
- Python_Cylindrical_sponge_dP12(G)
- Python_Flat_sponge_dP(G)
- Python_Flat_sponge_dP(Qs)
- Python_Flat_sponge_dP12(G)

These programs also take the specimen porosity ε into account, which renormalizes the filtration velocity $u = \frac{Q}{A\varepsilon}$.

The work of Wei Zhong et al. contains detailed experimental data for flat porous specimens measured under free-discharge conditions, which corresponds to equation (40). In their work, the coefficient β is dimensionless; therefore, for use in our equations it must be divided by $\sqrt{k}$.

Table 2. Properties of the test porous media in this study.

Porous Media	Estimated Particle Diameter (μm)	Length (mm)	Diameter (mm)	Porosity
Group 1	40–50 μm	3	7, 8, 9, 10, 11	0.395, 0.392, 0.388, 0.389, 0.397
Group 2	55–65 μm	3	7, 8, 9, 10, 11	0.418, 0.419, 0.418, 0.418, 0.416
Group 3	80–90 μm	3	7, 8, 9, 10, 11	0.420, 0.412, 0.413, 0.420, 0.425

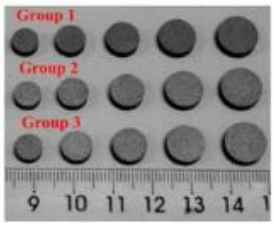


Table 3. Permeability and inertia coefficients of the porous media by the steady-steady method.

Porous Medium	Group 1		Group 2		Group 3	
	K (10^{-12} m^2)	β	K (10^{-12} m^2)	β	K (10^{-12} m^2)	β
7 mm	2.15	0.512	3.73	0.522	5.02	0.608
8 mm	2.02	0.515	3.64	0.585	4.99	0.617
9 mm	2.02	0.486	3.65	0.613	4.86	0.588
10 mm	2.048	0.467	3.68	0.578	5.05	0.598
11 mm	2.095	0.491	3.55	0.623	4.93	0.615

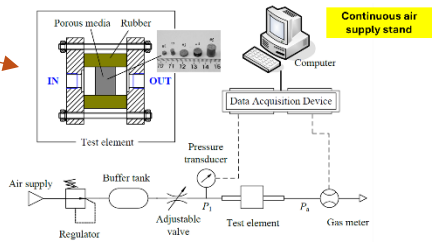
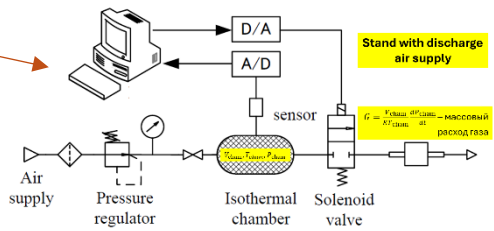


Table 4. Results of fitting coefficients K and β by the discharge method.

Porous Medium	Group 1		Group 2		Group 3	
	K (10^{-12} m^2)	β	K (10^{-12} m^2)	β	K (10^{-12} m^2)	β
7 mm	1.96	0.411	3.71	0.562	5.03	0.468
8 mm	1.97	0.467	3.68	0.641	4.91	0.537
9 mm	2.12	0.509	3.63	0.658	4.98	0.582
10 mm	1.98	0.467	3.66	0.621	4.95	0.562
11 mm	2.08	0.574	3.54	0.739	5.01	0.621



Let us select some representative values from these tables to estimate the pressure drop in our cylindrical sponge, for example (screenshots of the program console from the **Python_Cylindrical_sponge_dP(Qs)** folder):

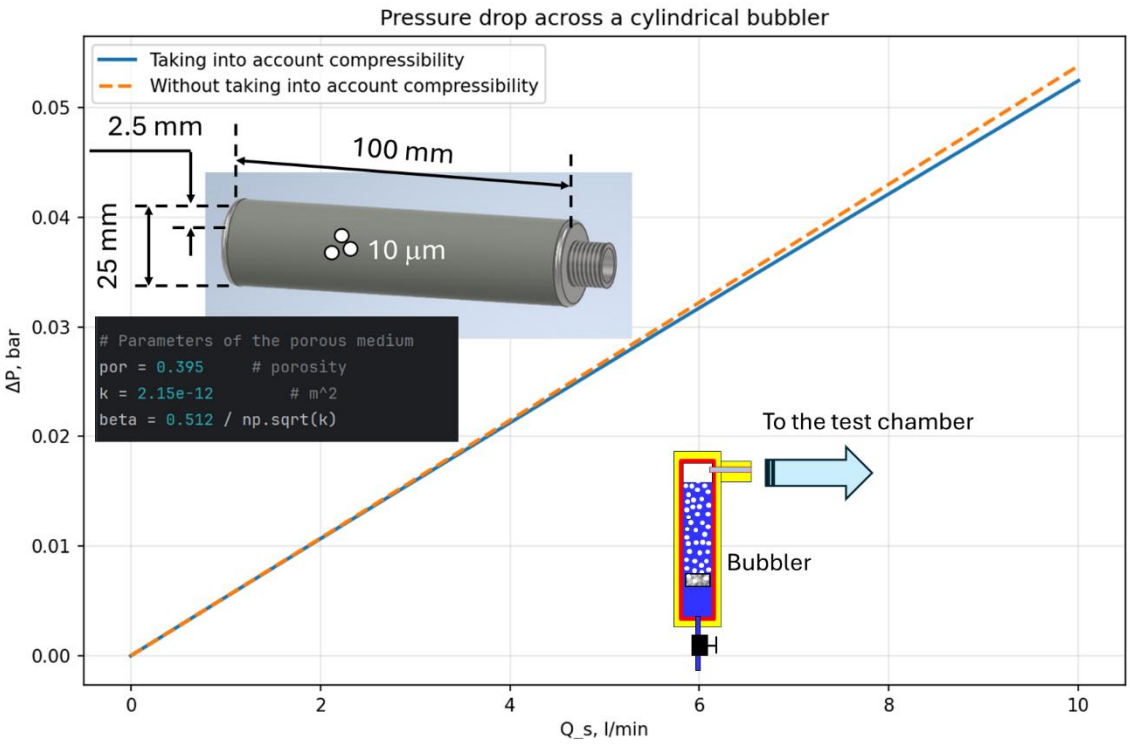
```
# Parameters of the porous medium
por = 0.395     # porosity
k = 2.15e-12        # m^2
beta = 0.512 / np.sqrt(k)
```

Остальные параметры, используемые при расчёте:

```
# Free flow range after bubbler, l/min
Q_min_lpm = 0.0
Q_max_lpm = 10.0
n_points = 300

# Bubbler geometry
R_out_mm = 12.5     # mm
R_in_mm = 5.0      # mm
L_c_mm = 100.0      # mm
```

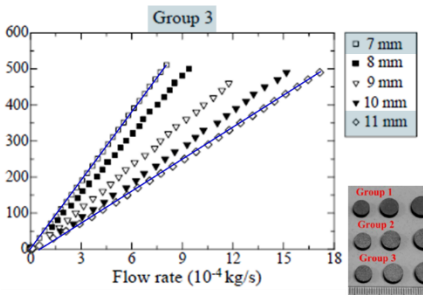
```
# Gas temperature in the sponge
T_s_C = 30.0        # degrees Celsius

# Outlet pressure
P_atm = 101325.0    # Pa (absolute)

# Газ: сухой воздух
mu_g = 1.872e-5     # Pa*s, dynamic viscosity of dry air at ~30 C
R = 287.05          # J/(kg*K), specific gas constant of air
```

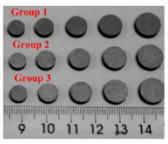
В результате получаем следующие зависимости (без/с учетом сжимаемости), рассчитанные по формулам (21) и (39):



Thus, the maximum pressure drop across our bubbler (tested with dry air without immersion in water) should be approximately 5 kPa. Moreover, as the graphs show, compressibility is still insignificant at flow rates of up to 10 litres per minute. The resulting curves categorically diverge from those provided by Saifilter.

To further verify our calculations, we reproduced the $\Delta P_{\text{flat}}(G)$ curve from Wei Zhong et al., who measured the pressure drop across flat samples ("bits") as a function of mass flow rate. Equation (40) with Group 3 parameters for two diameters, 7 mm and 11 mm, was used for the calculations. Our graphs and those from the paper completely matched, as shown in the figure below. The observed large pressure drop values are due to the small area of the "bits." Our cylindrical sponges have an area tens of times larger, so the pressure drop should be on the order of a few kPa, not fractions of a bar, as in the Saifilter graphs. "Parabolic curves" over a wide range of gas mass flow rates, which inflate the pressure drop values, only occur if gas compressibility is not taken into account.





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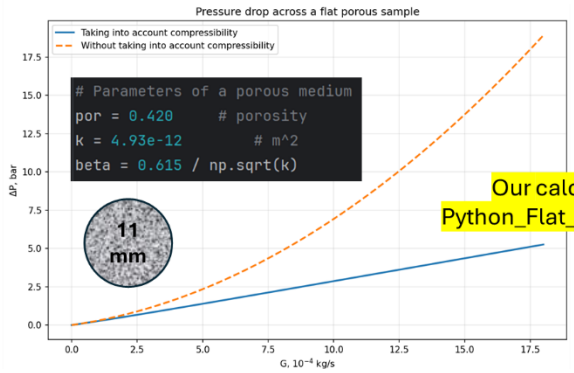
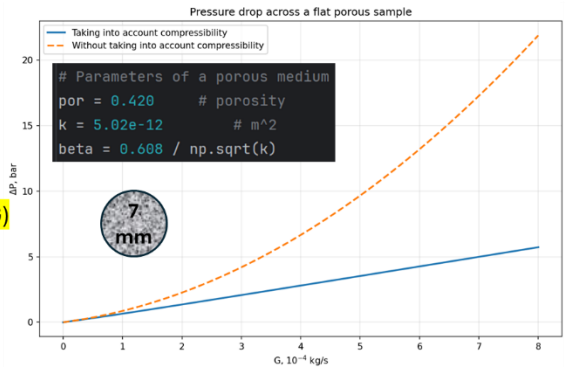
Our calculations:
Python_Flat_sponge_dP(G)